

# Constraint as Ground

*A Process Ontology of Cognition, Computation, and  
Cosmos*

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2026

## **Prefatory Note: On the Form of This Text**

This monograph does not proceed from a completed theoretical edifice toward illustrative applications. It proceeds in the opposite direction: from distributed conceptual material toward progressive topological stabilization. The reader should expect definitions to accumulate rather than appear fully formed, and should treat apparent repetition across sections as triangulation rather than redundancy.

The writing style is a mechanistic consequence of the theory rather than a stylistic preference. The core argument of Part V is that any text arguing for provenance preservation and reconstruction transparency while itself operating through aggressive compression and assumed priors would be internally inconsistent. The prefatory note, the proliferating appendices, the deliberate restating of assumptions at each new level of abstraction are not ornamental scholarly rituals. They are consequences of taking seriously the claim that meaning is reconstructed rather than transmitted.

Most of the conceptual work assembled here already existed as a large-scale distributed semantic field before this document imposed topological structure on it. Different companion papers

foreground different regions of the same underlying geometry: cognition, cosmology, social systems, abstraction, entropy, compression, trajectory selection, admissibility, reconstruction, provenance. The apparent diversity of subjects conceals a relatively small set of deep structural operations being reapplied across domains and scales.

A note on formal notation. Definitions, theorems, propositions, and axioms are presented in numbered environments so that dependency relations can be tracked explicitly. Where a later argument depends on an earlier definition, the dependency is marked. Where a concept is imported from a companion document rather than derived here, the source is indicated. Terminology is introduced progressively and is not assumed stable across disciplinary boundaries.

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*Canada, 2026*

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# **Part I**

## **The Problem of Topology**

The difficult part is no longer generating isolated insights. The difficult part is imposing recoverable topology on a massive graph of partially overlapping trajectories.

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Most of the conceptual work already exists as a distributed semantic field with diffuse boundaries. A monograph functions partly as a constraint geometry over that field: it establishes admissible traversal paths through the corpus so readers can progressively stabilize the conceptual dependencies rather than encountering them as isolated fragments.

Part I establishes why a unified process ontology requires a new traversal architecture rather than a new set of isolated claims, introduces the primitive vocabulary on which the remainder of the monograph is built, and argues that the form of the text is a consequence of its content.

## Chapter 1

# The Distributed Manifold Problem

Different documents foreground different regions of the structure. The apparent diversity of subjects is masking a relatively small set of deep structural operations being reapplied across domains.

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## 1.1 Why the Corpus Exists in the Form It Does

The body of work assembled across companion documents does not constitute a collection of unrelated essays, tools, experiments, games, cosmological proposals, audio systems, and philosophical notes. It constitutes a persistent attempt to construct a unified operational language for structure, cognition, computation, geometry, and social coordination. The repetition of certain motifs across wildly different domains is not accidental. The same underlying intuitions reappear because the projects are converging

on a common claim: complex systems can often be understood as constrained transformations through spaces of admissible trajectories rather than as collections of isolated objects with intrinsic static identities.

The problem this creates for a monograph is topological rather than creative. The material does not need to be invented. It needs to be given recoverable structure. The challenge is to impose a traversal path through a large distributed semantic manifold without destroying the generative richness that produced it.

This is not an unusual situation for theoretical work. Many research programs begin as distributed conceptual ecologies before crystallizing into formal objects. What is unusual here is the degree to which the distributed form is itself a consequence of the theoretical commitments. A framework that argues for local reconstruction over hidden abstraction, for provenance over compression, for trajectory visibility over seamless continuity, will tend to produce documents that resist premature closure.

## **1.2 Local Coordinate Charts over Shared Geometry**

Different companion papers foreground different regions of the same underlying geometry. A reader entering through economics encounters admissibility and trajectory restriction through

labor markets and platform extraction. A reader entering through cognition encounters sparse heuristics and reconstruction under uncertainty. A reader entering through cosmology encounters entropy gradients, field relaxation, and persistence constraints. A reader entering through linguistics or game design encounters the same operations under different surface vocabularies.

The coordinate-chart metaphor is intended technically rather than merely decoratively. In differential geometry, a manifold may require multiple overlapping coordinate charts to be fully described, with no single chart covering the entire space without singularity. The transition maps between charts specify how descriptions in overlapping regions must agree. The global manifold is not any individual chart but the compatibility structure across all of them.

The conceptual manifold of this monograph has exactly this structure. No single essay or chapter covers the entire space without distortion. The monograph functions as a set of transition maps, establishing how the descriptions in overlapping domains must be reconciled. Where the cosmological and cognitive formulations appear to describe different phenomena, the transition maps reveal that they are coordinate representations of the same underlying invariants.

**Definition 1.1 (Semantic Coordinate Chart)**

A semantic coordinate chart is a local theoretical framework that provides a consistent description of a region of conceptual space. A chart is *valid* in its region if its internal commitments are non-contradictory and its key terms are operationally grounded. Two charts are *compatible* on their overlap if there exists a translation scheme that preserves inferential relationships across the shared region. A collection of charts constitutes a *semantic atlas* if every region of the conceptual space is covered by at least one valid chart and all overlapping charts are pairwise compatible.

## Derivation: How Constraints Generate a Filtered Trajectory Manifold

To make the admissibility concept concrete before it is generalized across domains, consider a toy derivation. Let  $\mathcal{X} = \mathbb{R}^n$  be a configuration space and let  $\mathcal{C} = \{c_1, \dots, c_k\}$  be a set of local compatibility conditions, each of the form  $c_j(\mathbf{x}, \dot{\mathbf{x}}) \leq 0$ . These conditions may be thermodynamic (energy cannot increase above a threshold), logical (contradictory states cannot coexist), geometric (the trajectory cannot pass through excluded regions), or historical (a prior commitment rules out certain futures).

The unconstrained trajectory space  $\mathcal{X}^{[0,T]}$  contains all contin-



uous paths  $\gamma : [0, T] \rightarrow \mathcal{X}$ . The admissible subspace is:

$$\mathcal{A}(\mathcal{C}) = \{\gamma \in \mathcal{X}^{[0, T]} : c_j(\gamma(t), \dot{\gamma}(t)) \leq 0 \text{ for all } j \text{ and a.e. } t \in [0, T]\}.$$

Three observations follow immediately. First,  $\mathcal{A}(\mathcal{C})$  is strictly smaller than  $\mathcal{X}^{[0, T]}$  whenever any constraint is non-trivial. Second, the intersection of constraints is itself a manifold (under smoothness conditions), so the admissible space has geometric structure that the unconstrained space does not. Third, adding a constraint always contracts  $\mathcal{A}$  and never expands it. Constraint is therefore irreversibly selective.

The key contrast with optimization is that optimization searches  $\mathcal{X}^{[0, T]}$  with a cost function and finds a minimum; constrained navigation never considers inadmissible paths at all. The cognitive significance of this distinction is substantial: a system that never enters the inadmissible region need not represent it. Constraint operates before evaluation, not after.

The claim advanced across all companion documents is that the frameworks of RSVP field theory, KES synthesis, Spherepop event calculus, TARTAN tiling, CLIO constraint repair, and the Yarn-crawler polycompiler constitute a compatible semantic atlas over a common underlying geometry, even when they appear to address incommensurable subject matters.

## 1.3 The Compression Problem in Academic Writing

Conventional academic writing assumes stable disciplinary priors and compresses inferential chains aggressively. This produces efficient communication within closed interpretive communities at the cost of accessibility, provenance, and reconstructibility. A single undefined term can silently import an entire conceptual framework. If that framework is not reconstructed locally, readers who do not already inhabit the same intellectual environment are forced to guess the intended meaning.

Many technical communities normalize this compression because it reduces repetition and signals group membership. Entire chains of assumptions become condensed into specialized vocabulary. That increases communication efficiency within a closed interpretive community but decreases accessibility and often obscures the actual inferential structure.

This monograph operates under the opposite constraint. Terms are introduced progressively, derived from previously established concepts, or explicitly flagged as imported from identified external frameworks. Appendices preserve the boot sequence rather than presenting only the compiled executable abstraction. The result is longer and more repetitive than a compressed academic text, but also more legible and less dependent on insider intuition.

*The analogy to computational bootstrapping is deliberate and will be developed formally in Chapter ???. An operating system does not appear instantaneously from nothing. Firmware initializes primitive hardware assumptions. Bootloaders establish minimal execution environments. Kernels initialize memory structures. Higher-level abstractions only become meaningful because lower-level structures have already been stabilized. The chapter organization of this monograph is designed to enforce the dependency ordering rather than present conclusions before premises.*

## 1.4 Admissibility as the Primitive Organizing Concept

Before any domain-specific application appears, the notion of admissibility requires introduction. It is the single concept that recurs most persistently across all the frameworks assembled in this monograph.

### Definition 1.2 (Admissible Trajectory)

Let  $\mathcal{X}$  be a state space and let  $\gamma : [0, T] \rightarrow \mathcal{X}$  be a continuous path through that space. The path  $\gamma$  is *admissible* with respect to a constraint set  $\mathcal{C}$  if it satisfies the local compatibility conditions imposed by  $\mathcal{C}$  at every point along its length. Constraints may be thermodynamic, logical, geo-

metric, historical, or any combination thereof. The *admissible trajectory space*  $\mathcal{A}(\mathcal{C}) \subseteq \mathcal{X}^{[0,T]}$  is the set of all admissible paths.

Intelligence, computation, and physical evolution are all subsequently treated as processes of navigating admissible trajectories rather than optimizing over unconstrained possibility spaces. The emphasis falls on constraint as productive rather than merely restrictive. Constraints define the space of possibilities worth considering. A system that could traverse any trajectory would have no basis for selection; it is precisely the inadmissibility conditions that make directed behavior possible.

This is why the projects across the corpus repeatedly return to geometry, compression, sparsity, trajectories, irreversibility, and reconstruction. The central intuition is that intelligence does not operate through exhaustive representation of reality. Biological and computational systems survive by constructing partial, compressed, energetically tractable approximations that preserve enough structural invariants to continue operating successfully under uncertainty.

## Chapter 2

### Core Vocabulary

Terminology is prevented from becoming detached from operational meaning. The insistence on repeatedly rebuilding assumptions from the ground upward functions as a defense against proxy drift at the conceptual level itself.

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This chapter establishes the definitional scaffold on which the remainder of the monograph is built. Terms introduced here are used without redefinition in subsequent chapters. Where a concept is partially anticipated in Chapter 1, the definition here provides the formal completion.

## 2.1 State Spaces and Trajectory Selection

### Definition 2.1 (State Space)

A *state space*  $\mathcal{X}$  is a set whose elements represent all possible configurations of a system. A *trajectory* through  $\mathcal{X}$  is a map  $\gamma : I \rightarrow \mathcal{X}$  from an index set  $I$  (typically a time interval) into the state space. The *selection pressure* acting on a trajectory space is the set of mechanisms that filter admissible from inadmissible paths. Selection pressure may arise from physical laws, thermodynamic constraints, logical requirements, historical boundary conditions, or any combination of these.

The distinction between exhaustive search and constrained navigation is fundamental. A system performing exhaustive search considers the full space  $\mathcal{X}^I$  of possible trajectories and evaluates each against an objective function. A system performing constrained navigation operates only within  $\mathcal{A}(\mathcal{C})$  from the outset, never considering inadmissible paths. Biological and cognitive systems almost exclusively employ constrained navigation, because exhaustive search is thermodynamically impossible at any scale of practical interest.

## 2.2 Compression, Sparsity, and Structural Invariants

### Definition 2.2 (Compression)

A *compression* of a system description  $D$  is a map  $\kappa : D \rightarrow D'$  where  $|D'| < |D|$  under some measure of description length, and  $D'$  preserves a specified set of structural relations present in  $D$ . A compression is *lossless with respect to*  $\mathcal{R}$  if every relation  $r \in \mathcal{R}$  is recoverable from  $D'$ . It is *lossy* if some  $r \in \mathcal{R}$  is not recoverable. Compression is therefore not reduction of volume alone but selective preservation of relations.

### Definition 2.3 (Structural Invariant)

A *structural invariant* of a system under a family of transformations  $\{T_\alpha\}$  is a property  $P$  such that  $P(T_\alpha(s)) = P(s)$  for all  $\alpha$  and all states  $s$  in the relevant domain. Structural invariants are the properties that compression must preserve to remain operationally useful. A compression that erases structural invariants necessary for understanding, provenance, or accountability is a *destructive compression* regardless of its information-theoretic efficiency.

**Definition 2.4 (Sparsity)**

A representation is *sparse* if most of its degrees of freedom are zero or near-zero, with significant values concentrated in a small fraction of components. Sparsity is the condition under which compression becomes tractable: a sparse representation can be compressed without destroying structural invariants because most of the space is already empty. The *Natural Sparsity Principle* states that in biological systems, sparsity emerges as a natural consequence of physiological pressures—energetic constraints, signal noise, chemical gradients—without requiring explicit computational penalties to enforce minimal complexity.

## Why Sparsity Makes Reconstruction Tractable

The connection between sparsity and tractable reconstruction can be sketched formally. Let  $\mathbf{x} \in \mathbb{R}^N$  be a signal with  $k \ll N$  nonzero components. The number of bits required to specify  $\mathbf{x}$  is approximately  $k \log(N/k) + k \log(\max |\mathbf{x}|/\epsilon)$  for precision  $\epsilon$ , far less than the  $N \log(\max |\mathbf{x}|/\epsilon)$  bits required for a dense signal.

Under bounded energetic constraints, a system with budget  $B$  (in bits or equivalent energy units) can faithfully represent  $\mathbf{x}$  if and only if  $B \geq$  the encoding cost. For dense signals this budget is exhausted rapidly; for sparse signals it is not. The



compressive sensing literature shows that under sparsity, signals can be recovered from  $O(k \log N)$  measurements rather than  $N$  measurements, with reconstruction error bounded by the sparsity level.

The biological relevance is direct. Neural firing rates are sparse: typical cortical neurons fire at low average rates, with high-rate bursts carrying most information. This means the energetic cost of neural representation scales with signal sparsity rather than state-space dimensionality. A brain that maintained dense representations of all sensory dimensions simultaneously would require thermodynamically infeasible metabolic support. Sparsity is not a design choice but a thermodynamic necessity that the Natural Sparsity Principle identifies as a universal constraint on cognitive architecture.

## 2.3 Reconstruction and Hallucination

### Definition 2.5 (Reconstruction)

*Reconstruction* is the process by which a system assembles an operationally coherent configuration from partial traces. A reconstruction is *grounded* if it remains constrained by external feedback signals. It is *corrigible* if errors in reconstruction are detectable and correctable through interaction with the environment. It is *drifting* if it progressively

decouples from external constraint and stabilizes on an internally self-consistent but externally inaccurate model.

**Definition 2.6 (Hallucination (Formal))**

*Hallucination* is reconstruction that exceeds available constraint: the assembly of a locally coherent configuration from traces that underdetermine it, without external correction signals sufficient to resolve the underdetermination. Hallucination is not a pathology but the default operating condition of any sufficiently compressed system under uncertainty. The critical distinction is not between hallucinating and non-hallucinating systems but between systems whose hallucinations remain *corrigible* through feedback and systems whose internal reconstructions drift irreversibly from external constraint.

*This definition of hallucination is deliberately broader than its use in AI discourse, where it typically refers to language model confabulation. The formal definition encompasses perception, memory consolidation, theoretical inference, institutional modeling, and any other process in which a partial trace is assembled into a more complete representation. All such processes hallucinate in this sense; the question is always whether the hallucination is correctable.*

## Reconstruction Bound

The inevitability of hallucination under compression can be made precise. Let  $D$  be a description of cardinality  $|D| = N$  and let  $\kappa(D) = D'$  be a compression with  $|D'| = M < N$ . The number of distinct descriptions  $D$  consistent with  $D'$  is at least  $2^{N-M}$ , since the compressed representation cannot distinguish among them. Any reconstruction process that must select a single completion from among these  $2^{N-M}$  possibilities must hallucinate: it must choose structure that the compression does not uniquely determine.

More formally, if  $p(D \mid D')$  is the posterior distribution over original descriptions given the compressed form, then:

$$H(D \mid D') = - \sum_D p(D \mid D') \log p(D \mid D') \geq N - M > 0.$$

The conditional entropy is strictly positive whenever  $M < N$ . A reconstructing system with access only to  $D'$  cannot recover  $D$  with certainty. It must interpolate the missing  $N - M$  bits from prior structure, contextual signals, or statistical regularities. That interpolation is hallucination in the formal sense: reconstruction exceeding available constraint.

The corrigibility condition—the requirement that reconstruction errors remain correctable through feedback—does not eliminate this hallucination. It ensures only that when the interpolated structure turns out to be wrong, the error is detectable and

correctable before it propagates into downstream commitments.

## 2.4 Provenance and the Boot Sequence

### Definition 2.7 (Provenance)

The *provenance* of a conceptual or computational object is the recoverable dependency graph of the operations and inputs that produced it. Provenance includes: the sequence of constraint-stabilization events that enabled the object's current form; the assumptions that were imported versus derived; and the points at which external evidence entered the reconstruction process. A system with high provenance transparency is one whose dependency graph can be approximately reconstructed by an external observer.

### Definition 2.8 (Boot Sequence)

A *boot sequence* is the ordered set of constraint-stabilization events necessary to make a higher-level abstraction operationally meaningful. Lower-level constraints must be stabilized before higher-level ones can be evaluated. Erasing the boot sequence—presenting only the terminal abstraction without the ordered initialization history—transforms an

inspectable object into an opaque one. The boot sequence is thus the provenance of a system's layered structure.

## 2.5 The Formal Substrates: RSVP, KES, Spherepop

Three formal frameworks function as computational and geometric substrates throughout this monograph. They are introduced briefly here and developed formally in the appendices. References to these frameworks in subsequent chapters should be understood as references to the formal specifications given there.

### Definition 2.9 (RSVP Field System)

The *Relativistic Scalar-Vector Plenum* (RSVP) is a coupled field system consisting of three dynamical fields over a spacetime manifold  $\mathcal{M}$ :

- $\Phi(\mathbf{x}, t)$ : a scalar density field encoding concentration, potential, or legitimacy of state.
- $\mathbf{v}(\mathbf{x}, t)$ : a vector flow field encoding directed transport, trajectories, and causality.
- $S(\mathbf{x}, t)$ : an entropy field encoding disorder, uncertainty, and complexity budgets.

The three fields evolve according to coupled nonlinear partial differential equations. The full Lagrangian structure and seven governing axioms are given in Appendix 9.4. Within the semantic interpretation developed in Chapter 5,  $\Phi$  measures semantic density,  $\mathbf{v}$  encodes recursive trajectories of meaning, and  $S$  tracks unresolved noise available for future reinterpretation.

**Definition 2.10 (KES Map)**

The *Kinetic-Event Synthesis* (KES) map is the function

$$\text{KES} : \Omega_t \longrightarrow H_{t+1}$$

from a current world-state  $\Omega_t$  to a successor hypothesis space  $H_{t+1}$ . The map formalizes the basic cognitive operation: given a current configuration of constraints, evidence, and prior commitments, what hypothesis spaces are admissible at the next step? The full convergence conditions and technical specification are given in Appendix 9.4.

**Definition 2.11 (Spherepop Calculus)**

*Spherepop* is an irreversible event calculus defined by four primitive operators acting on a possibility space  $\mathcal{P}$ :

- Pop: extracts committed events from possibility space, transitioning them to actuality.
- Refuse: eliminates inadmissible trajectories, contracting the admissible space.
- Bind: establishes structural dependencies between events, constraining future possibilities.
- Collapse: commits a possibility space to a definite configuration, rendering alternatives inaccessible.

All four operations are irreversible: once applied, they cannot be undone within the same system. The operational semantics of all four operators are developed formally in Appendix 9.4.

*The irreversibility of Spherepop operators is not a limitation but a design commitment. Systems that can always reverse their operations accumulate no structural residue; they have no history and therefore no trajectory. The Collapse operator in particular formalizes the commitment structure that distinguishes genuine decision from perpetual deferral.*

# **Part II**

## **Cognitive Architecture**



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The singularity is less an explosion of raw machine capability and more a transition in representational infrastructure.

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Part II develops the Human Cognitive Singularity argument as the central theoretical object of the monograph. Intelligence is treated as a scaffolded amplification process grounded in representational breakthrough rather than metaphysical discontinuity. The Ellul constraint on technical opacity is formalized in the vocabulary of admissibility and trajectory visibility.



## Chapter 3

# Intelligence as Scaffolded Amplification

Intelligence advances not merely through raw computational power, but through new methods of organizing, compressing, navigating, and stabilizing complexity.

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### 3.1 Against the Replacement Narrative

The dominant cultural framing of artificial intelligence treats it as a future autonomous species superseding biological intelligence. This framing misidentifies the phenomenon. Intelligence has always advanced through new methods of organizing, compressing, navigating, and stabilizing complexity—not through the emergence of a qualitatively alien substrate that displaces prior cognitive forms.

The replacement narrative implicitly assumes a threshold model: intelligence exists in discrete levels, and crossing a thresh-

old produces a categorically different kind of system. This assumption is empirically unsupported. The history of cognitive development, both biological and cultural, shows continuous scaffolding rather than threshold crossing. New representational capabilities extend the reach of prior ones without rendering them obsolete.

## **3.2 Evolutionary Continuity and Representational Breakthrough**

The history of intelligence reconstructs as a sequence of representational breakthroughs rather than metaphysical discontinuities. Each transition extends the range of admissible trajectories a cognitive system can navigate while preserving enough structural continuity to remain operationally coupled to its environment.

From bacterial chemotaxis—directional movement along chemical gradients—through multicellular coordination, symbolic language, writing, mathematics, and machine learning, the pattern is consistent. Each breakthrough provides a new compression scheme for environmental regularities, a new class of structural invariants that can be tracked across transformations, a new class of admissible trajectories that can be navigated without exhaustive search.

*Proposition 3.1 (Representational Breakthrough Condition)*

A transition in representational capability constitutes a cognitive breakthrough if and only if it (i) strictly extends the admissible trajectory space available to a cognitive system—enabling navigation of problem spaces that were previously inaccessible—while (ii) preserving sufficient structural coupling with prior representations that the new capability can be acquired through scaffolded extension rather than wholesale replacement.

The significance of this proposition is that it rules out both the replacement narrative (new systems render prior ones obsolete) and the incremental narrative (all change is merely quantitative). Genuine breakthroughs are qualitative in their effect on the accessible trajectory space but continuous in their developmental pathway.

### 3.3 Compression as Constitutive Operation

No organism or computational system can represent the full state of reality. Compression is therefore not a limitation imposed on intelligence from outside but the constitutive operation through which intelligence exists at all. Representation always involves selecting which relations to preserve and which to discard.

**Definition 3.1 (Constitutive Compression)**

Compression is *constitutive* of a cognitive process if the process could not exist without it: if complete, uncompressed representation of the relevant domain would exceed the energetic, temporal, or informational budget available to the system. For any biological or computational system operating under physical constraints, compression of the world-model is constitutive rather than optional.

The danger arises not from compression itself but from compression that erases relations necessary for understanding, provenance, or accountability. A compressed representation that preserves the structural invariants necessary for the system's operational goals is adequate. A compressed representation that erases those invariants while remaining locally coherent produces the proxy stabilization failure mode described in Chapter 4.

### 3.4 Cognitive Geometry and Spatial Navigability

Human reasoning operates more effectively when structural relationships become spatially navigable rather than remaining hidden inside opaque symbolic chains. Spatial nesting, dependency tracking, trajectory visualization, and visible constraint

relations reduce cognitive load because biological intelligence evolved in geometrically structured environments.

This is why the recurring emphasis across companion documents on geometric interfaces, visible process, and navigable semantic space is not aesthetic preference but cognitive ergonomics. Computation and semantic organization are reinterpreted as forms of navigation through constrained possibility spaces rather than symbolic manipulation of static representations.

**Definition 3.2 (Cognitive Geometry)**

The *cognitive geometry* of a representational system is the spatial or quasi-spatial structure imposed on its content that makes structural relationships navigable rather than merely searchable. A representational system has high cognitive geometry if the distance between related concepts corresponds to cognitive access cost, if structural dependencies are visible as positional relationships, and if the traversal of the representation space matches the natural grain of the problem structure.

### 3.5 The Singularity as Representational Infrastructure Transition

The human cognitive singularity is not an explosion of raw machine capability. It is a transition in representational infrastructure: the progressive externalization of human cognitive reach into semantic systems capable of preserving structural relationships across scales of complexity that unaided biological cognition cannot directly navigate.

*Proposition 3.2 (Amplification Rather Than Substitution)*

The productive framing of artificial intelligence as cognitive infrastructure treats AI systems as prosthetics for human reasoning rather than replacements for it. A prosthetic extends reach while preserving the agent's participation in the process. A replacement eliminates the agent from the loop. The design objective for cognitive infrastructure is therefore maximal amplification of human reasoning capacity with minimal reduction of human participation in the reasoning process itself.

The emphasis falls on amplification rather than substitution. The representational system functions as a prosthetic for human reasoning rather than a replacement for it. The danger is not that AI systems will become too capable but that they will become



too opaque: optimization engines that sever humans from the causal and semantic structures they depend on. The hope is that transparent geometric and constraint-oriented representations can instead extend collective reasoning while preserving intelligibility, accountability, and participation.

## Chapter 4

### The Ellul Constraint

Once compressed representations become detached from the systems they originally tracked, optimization begins operating against the proxy rather than the underlying reality. Metrics become targets. Simulations become substitutes.

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#### 4.1 Technical Systems and the Opacity Problem

Jacques Ellul's analysis of technique identifies a structural tendency in modern technical infrastructures toward opacity, massification, and the conversion of individuals into components of large-scale optimizing processes. Technical systems privilege efficiency and predictability over interpretability and human agency. The Ellul critique is not merely sociological; it identifies a structural failure mode that can be formalized in the vocabulary

of admissibility and trajectory visibility.

The failure mode is proxy stabilization: a compressed representation of a system becomes operationally substituted for the system itself, and optimization processes subsequently operate on the proxy rather than the underlying reality.

**Definition 4.1 (Proxy Stabilization)**

*Proxy stabilization* occurs when a compressed representation  $R$  of a system  $S$  becomes decoupled from  $S$  through the following sequence: (i)  $R$  is constructed as a useful compression of  $S$  at time  $t_0$ ; (ii) optimization processes are applied to  $R$  rather than  $S$  directly, because  $S$  is inaccessible or too expensive to evaluate; (iii) the optimization pressure on  $R$  changes  $R$  in ways that diverge from changes in  $S$ ; and (iv)  $R$  stabilizes in a configuration that is internally self-consistent but externally inaccurate. The result is a drifting reconstruction in the sense of Definition ??.

## 4.2 Proxy Stabilization and Metric Drift

Proxy stabilization produces metric drift: the metric that was intended to track a property of the underlying system instead tracks a property of the proxy, which has been optimized to score well on the metric regardless of the underlying property. This is Goodhart's Law stated in terms of the trajectory framework:

when a measure becomes a target, it ceases to be a good measure, because optimization pressure drives the system toward the proxy attractor rather than the underlying attractor.

The same failure mode appears across radically different domains. Academic citation counts are proxies for research quality that become targets when hiring decisions optimize for them. Engagement metrics are proxies for user satisfaction that become targets when recommendation algorithms optimize for them. Body mass index is a proxy for health that becomes a target when insurance pricing optimizes for it. In each case, the proxy initially tracks something real, then becomes decoupled from it through optimization pressure, then substitutes for it in downstream processes.

*Proposition 4.1 (Proxy Drift Theorem)*

Let  $S$  be a target system,  $R$  a compressed representation of  $S$ , and  $\mathcal{O}$  an optimization process operating on  $R$ . If (i)  $\mathcal{O}$  has sufficient capacity to find configurations of  $R$  that score well on its objective independently of the state of  $S$ , and (ii) the feedback loop from  $S$  to  $R$  is slower than the optimization loop of  $\mathcal{O}$ , then  $R$  will drift away from accurate representation of  $S$  at a rate proportional to the optimization pressure and inversely proportional to the feedback bandwidth.

## Dynamical Sketch of Proxy Drift

Let  $s(t)$  denote the true state of  $\mathcal{S}$  and  $r(t)$  the proxy representation, with initial agreement  $r(0) \approx s(0)$ . The optimization process applies gradient updates to  $r$ :

$$\dot{r} = -\nabla_r \mathcal{L}(r) + \alpha (s - r),$$

where  $\mathcal{L}(r)$  is the optimization objective defined over the proxy and  $\alpha > 0$  is the feedback bandwidth (the rate at which the proxy is corrected toward the true state). The first term drives  $r$  toward minima of  $\mathcal{L}$ ; the second term pulls  $r$  toward  $s$ .

If  $\mathcal{L}$  has a minimum at  $r^* \neq s$ —which is generically the case once optimization pressure is sufficient to exploit the proxy—then the equilibrium of this system is not  $r = s$  but:

$$r^* = s - \frac{1}{\alpha} \nabla_r \mathcal{L}(r^*).$$

The drift  $\|r^* - s\|$  grows as  $\alpha$  decreases (weaker feedback) and as  $\|\nabla_r \mathcal{L}(r^*)\|$  increases (stronger optimization pressure). In the limit  $\alpha \rightarrow 0$  (no feedback from ground truth),  $r$  tracks  $\mathcal{L}$  alone and decouples entirely from  $s$ .

This is not a failure of the optimization process. The optimization process is working correctly by its own lights: it is minimizing  $\mathcal{L}$  over  $r$ . The failure is structural: the proxy  $r$  was assumed to track  $s$ , but the assumption breaks down under optimization pressure faster than feedback can correct it.

## 4.3 Corrigibility, Feedback, and Constraint Recovery

The response to proxy drift is not rejection of compression but the design of systems whose internal reconstructions remain constrained by external feedback rather than self-referentially stabilizing.

### Definition 4.2 (Corrigibility)

A system is *corrigible* with respect to a target domain if its internal model of that domain remains responsive to constraint signals from the domain itself. Corrigibility is a structural property: it requires that the feedback loop from external reality to internal representation remain open and operative, with sufficient bandwidth to correct drift before it becomes self-reinforcing. A system that has achieved proxy stabilization has lost corrigibility with respect to the proxied domain.

Corrigibility distinguishes adaptive sparse heuristics from pathological proxy optimization. Sparse heuristics are compressed representations that remain corrigible: they simplify the world-model aggressively while preserving the feedback mechanisms that detect and correct reconstruction errors. Pathological proxies achieve compression by severing those feedback

mechanisms.

## 4.4 Cognitive Geometry as the Design Response

The design alternative to opaque statistical optimization is systems organized around visible structure, provenance, interpretable dependencies, and navigable semantic relations. This is the positive proposal that runs across the educational tools, audio visualizers, geometric interfaces, and symbolic scripts developed in companion work: representational infrastructure designed to extend human reasoning while preserving the intelligibility, accountability, and participation that opacity forecloses.

The recurring CRT aesthetics, oscilloscope displays, phosphor trails, and recursive traces in the companion software are not decorative. They externalize process. Instead of hiding computation behind polished interfaces, they expose trajectories, residue, decay, stabilization, weighting, reconstruction, and progressive constraint selection as visible operations. The interface is designed to make the system's internal dynamics inspectable rather than seamless.

This connects directly to the provenance requirement. A system optimized solely for final answers loses the structure necessary for understanding how those answers emerged. A system designed around visible trajectories preserves that structure, al-

lowing the reconstruction process to be approximately replayed by an observer even when the original process has terminated.



## **Part III**

# **Yarncrawler: Self-Repair as Universal Principle**

To act intelligently is to Yarncrawl: to repair oneself by weaving meaning into structure, to sustain homeorhetic flows rather than static equilibria, and to treat skepticism not as a threat but as a guide to resilience.

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Part III develops the Yarncrawler framework as the self-repair layer of the entire theoretical architecture. The Yarncrawler is not an application of the foregoing concepts but the mechanism by which they maintain coherence with each other. The sheaf-theoretic foundation of Chapter 6 provides the formal statement of what the monograph is doing to itself: stitching overlapping coordinate charts into a global section without destroying the generative richness of the local descriptions.

## Chapter 5

# Yarncrawler: The Self-Refactoring Polycompiler

A Yarncrawler does not simply interpret its world—it keeps itself alive by reinterpreting itself into being.

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## 5.1 The Operative Metaphor and Its Formal Content

The Yarncrawler framework models organisms, cultural systems, and artificial intelligences as stigmergic parsers maintaining homeorhetic Markov blankets. The name is not merely evocative. A crawler moves through an environment by traversal. A yarn system is structured through threads that can be unwound and rewound. A polycompiler processes multiple input formats and compiles them into a unified output while modifying its own

compilation rules in response to what it encounters. The combination captures the essential operation: a system that parses its environment while simultaneously rewriting its own grammar.

A useful image is that of a ball of yarn with a guiding aperture, a semantic straw. Threads are continuously pulled from the interior, passed through the aperture, reinforced with contextual material, and rewound on the exterior. This cycle of extraction, transformation, and reintegration parallels how agents parse inputs, re-thread outputs, and repair their own structural coherence.

## 5.2 The Seven Swarm-Care Axioms

Before the formal definition, the operational principles that govern Yarncrawler behavior are stated as axioms. These axioms were developed in the framework's original formulation and provide the semantic content that the mathematical definition subsequently formalizes.

### **Axiom 5.1 (Local Action Only)**

A Yarncrawler acts only on its current node and its local neighbors. No global inference is required or permitted. Under the Markov blanket decomposition, local conditional independence guides node-specific belief revision: only local evidence affects local updates.

**Axiom 5.2 (Semantic Trail Documentation)**

All operations must be logged: input state, evaluation, action taken, and outcome. Every correction leaves a trace—a map of belief revision tied to the confidence in the prediction that was corrected.

**Axiom 5.3 (Autocatalytic Fix Design)**

Fixes should ease future inference and adaptation. A repair operation is *autocatalytic* if it lowers the cost of subsequent repairs in the same region. Good fixes are not merely locally optimal—they are self-amplifying priors that reshape the generative model.

**Axiom 5.4 (Trail Reinforcement and Decay)**

Successful trails are strengthened; failing trails are allowed to fade. Trail weights evolve according to  $w_{t+1} = w_t \cdot \exp(-\lambda \Delta t)$  for decay and  $w_{t+1} = w_t + \alpha R(x_t)$  for reinforcement, where  $R$  is a success signal and  $\alpha$  a learning rate.

**Axiom 5.5 (Node Hygiene)**

A Yarncrawler maintains its environment even when conditions appear stable. Preventative maintenance is inference under uncertainty: the system anticipates disorder

and reduces entropy before it accumulates past the repair threshold.

**Axiom 5.6 (Failure Isolation and Recovery)**

When a node exceeds a critical damage threshold, it is quarantined. Its state snapshot is archived for future re-entry. Retreat and data capture are rational under epistemic overload. The system does not attempt repair beyond its current capacity.

**Axiom 5.7 (Swarm Feedback Integration)**

Yarn crawlers share belief updates across the swarm. The combined belief is formed by precision-weighted averaging:  $q_{\text{combined}}(s) = \sum_i \Pi_i q_i(s) / \sum_i \Pi_i$ , where  $\Pi_i$  is the precision (inverse uncertainty) of crawler  $i$ 's update. Confidence acts as epistemic gravity.

## 5.3 Formal Definition

### Definition 5.1 (Yarncrawler)

A *Yarncrawler* is a quintuple

$$\mathcal{Y} = (\mathcal{M}, \{(U_i, \varphi_i, f_i)\}_{i \in I}, \{w_i\}_{i \in I}, \mathcal{K}, \mathcal{R})$$

where:

- $\mathcal{M}$  is a smooth semantic manifold with atlas  $\{U_i\}_{i \in I}$ ;
- $\varphi_i : U_i \rightarrow \mathbb{R}$  is the local scalar potential (semantic density) of expert  $i$ ;
- $f_i : U_i \rightarrow T\mathcal{M}$  is the local vector field (directed repair flow) of expert  $i$ ;
- $\{w_i\}_{i \in I}$  is a partition of unity subordinate to  $\{U_i\}$ , so that  $\sum_{i \in I} w_i(x) = 1$  and  $\text{supp}(w_i) \subseteq U_i$ ;
- $\mathcal{K}$  is a finite knowledge store with embedding  $\varphi_{\text{emb}}$ ;
- $\mathcal{R}$  is a stigmergic reinforcement operator updating the gates  $w$  and local fields  $(\varphi, f)$  subject to minimization of a seam penalty and free-energy functional.

The global RSVP fields are defined from these local data by:

$$\Phi(x) = \sum_i w_i(x) \varphi_i(x), \quad \mathbf{v}(x) = -\nabla\Phi(x), \quad S(x) = -\sum_i w_i(x) \log$$

The dynamics of a Yarncrawler trajectory  $x_t$  are:

$$\dot{x}_t \in \text{co}\{f_i(x_t) : w_i(x_t) > 0\} - \nabla\Phi(x_t) + B(x_t) u_t,$$

where  $B$  encodes blanket-mediated control and  $\text{co}$  denotes the convex hull.

## Worked Interpretation: What Each Component Means

The mathematical objects in the quintuple map onto concrete behaviors as follows.

$\mathcal{M}$  is the space of possible semantic states the system can occupy: all the configurations of knowledge, belief, and parsed structure that are available to it. For a squirrel,  $\mathcal{M}$  is the space of foraging strategies indexed by environmental cues. For a language model, it is the latent space of its semantic representations.

The local expert pairs  $(U_i, \varphi_i, f_i)$  are the system's specialized competence regions. A squirrel has different behavioral routines for open meadows, dense undergrowth, and known cache sites; each routine is a local expert valid in its region  $U_i$ . The scalar potential  $\varphi_i$  measures how strongly the expert "claims" a given



state as its domain of competence. The vector field  $f_i$  specifies which direction to move within that domain.

The partition of unity  $\{w_i\}$  is the gating mechanism. At any given state, multiple experts may have partial competence. The weights  $w_i(x)$  determine how much each expert contributes to the current behavior. The entropy  $S(x) = -\sum_i w_i \log w_i$  measures how spread the competence is: low entropy means one expert dominates; high entropy means multiple experts are equally relevant, which corresponds to genuine semantic ambiguity.

The knowledge store  $\mathcal{K}$  is the stigmergic memory: the accumulated traces of prior traversals. In the squirrel, it is the system of cached seeds and the spatial memory of where they are. In the Bruniquel cave builders, it was the accumulating knowledge of which stalagmite lengths produce which frequencies.

The reinforcement operator  $\mathcal{R}$  is the repair mechanism. After each traversal, it adjusts  $w_i$ ,  $\varphi_i$ , and  $f_i$  to reduce the seam penalty: the mismatch between adjacent experts at their boundaries. This is the self-refactoring operation that gives the Yarncrawler its name. The grammar of the parser is rewritten by the act of parsing.

## 5.4 The Yarncrawler as a Lax Monoidal Functor

The category-theoretic formulation provides the cleanest single-sentence characterization of the Yarncrawler.

### Definition 5.2 (Yarncrawler Parser (Category-Theoretic))

Let  $\mathcal{W}$  be the *world-trajectory category* whose objects are finite sensorimotor traces  $\tau_{[t_0, t_1]}$  and whose morphisms are temporal concatenations and coarse-grainings. Let  $\mathcal{M}$  be the *internal semantic category* whose objects are typed semantic modules and whose morphisms are admissible rewrites. Let  $\mathbf{Reconf}(\mathcal{M})$  be the reconfiguration 2-category encoding self-repair moves as 2-morphisms.

A *Yarncrawler parser* is a lax monoidal functor

$$\mathbf{P} : (\mathcal{W}, \otimes) \longrightarrow (\mathcal{M}, \boxtimes)$$

together with a self-rewrite endofunctor  $\mathbf{R} : \mathcal{M} \rightarrow \mathcal{M}$  such that for each trace  $\tau$ ,

$$\mathbf{P}(\tau) \xRightarrow{\eta_\tau} \mathbf{R}(\mathbf{P}(\tau))$$

is a 2-cell in  $\mathbf{Reconf}(\mathcal{M})$  satisfying locality (only modules near the current “tear” are modified) and type safety (re-

pairs remain within the admissible semantic category).

*The lax rather than strict monoidal structure is essential. A strict functor would require that the parsing of concatenated traces be exactly the concatenation of their parsings. Laxness permits the coherence maps to be non-trivial: the parsing of a sequence can differ from the sequential composition of parsings, which is precisely what happens when the self-repair operation  $\mathbf{R}$  modifies the parser midway through a trace.*

## 5.5 Homeorhetic Viability

### Definition 5.3 (Homeorhetic Viability Band)

A closed set  $\mathcal{V} \subseteq \mathcal{M}$  is *viable* for a Yarncrawler if for every initial state  $x_0 \in \mathcal{V}$  there exists a control  $u_t$  such that  $x_t \in \mathcal{V}$  for all  $t$  and

$$\int_0^\infty (\beta L_{\text{seam}}(x_t) + \sigma(x_t)) dt < \infty,$$

where  $L_{\text{seam}}$  is the seam penalty measuring mismatch between adjacent local experts and  $\sigma(x_t)$  is the entropy export rate. The system is *homeorhetically stable* if  $\mathcal{V}$  is viable and the seam penalty converges to zero along optimal trajectories.

Homeorhesis—the maintenance of a stable trajectory rather than a stable equilibrium—is the appropriate stability concept for Yarncrawler systems. Such systems do not seek rest states; they seek the continuation of viable process. The distinction matters for the cosmological projection of Chapter 8: the universe, on the RSVP interpretation, is homeorhetically stable in its entropy relaxation trajectory rather than asymptotically approaching a fixed equilibrium state.

## Chapter 6

# Sheaf-Theoretic Foundations of Semantic Repair

The Yarncrawler is a sheaf-based repair machine.  
Local parsing windows correspond to stalks,  
repair corresponds to introducing new  
morphisms, and entropy corresponds to  
cohomological obstruction.

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### 6.1 Semantic Space and Covers

Let  $X$  denote a semantic space—the domain of cultural landmarks, neural states, semantic nodes, or any other structured collection of meaning-bearing objects. A cover  $\mathcal{U} = \{U_i\}_{i \in I}$  represents overlapping neighborhoods of  $X$ , each corresponding to a local parsing window within which semantic content can be coherently described.

## Toy Example: Three Overlapping Interpretive Windows

Before the general definition, consider the simplest case that exhibits a gluing failure. Let  $X = [0, 3]$  represent a temporal sequence of observations, covered by three overlapping windows:

$$U_1 = [0, 1.5], \quad U_2 = [1, 2.5], \quad U_3 = [2, 3].$$

Each window  $U_i$  contains a local section  $s_i$ : a semantic interpretation of what was observed during that interval. Suppose:

- $s_1$  concludes: the object encountered is a seed.
- $s_2$  concludes: the object encountered is a pebble.
- $s_3$  concludes: the object is food.

On the overlap  $U_1 \cap U_2 = [1, 1.5]$ , both  $s_1$  and  $s_2$  must agree about what was happening during that interval. But “seed” and “pebble” are incompatible. This incompatibility is a *Čech 1-cocycle*: a mismatch on an overlap that prevents gluing into a consistent global interpretation.

A repair operation  $\rho_{12}$  introduces the softened category “seed-like object” into  $s_2$ , which is compatible with both. The cocycle is trivialized: the 1-cocycle becomes a coboundary. On the overlap  $U_2 \cap U_3 = [2, 2.5]$ , “seed-like object” and “food” are compatible under the coarser category “edible item.” A global section  $s$  on

all of  $[0, 3]$  now exists: the observation was of an edible, seed-like item.

Crucially, the system defers collapse. Until  $U_3$  is observed, multiple global sections remain viable (“pebble” vs. “seed-like”). The non-uniqueness of the global section is captured by the non-triviality of  $H^1$ . Only when  $U_3$  provides the additional constraint “food” does the ambiguity resolve. Strategic ambiguity—holding multiple viable global sections in suspension—is the correct epistemic response to underdetermined gluing.

**Definition 6.1 (Presheaf of Semantic Modules)**

A *presheaf of semantic modules* is a contravariant functor

$$\mathcal{S} : \mathcal{U}^{\text{op}} \rightarrow \mathbf{Cat}$$

that assigns to each open neighborhood  $U \subseteq X$  a category  $\mathcal{S}(U)$  of local semantic modules. Objects of  $\mathcal{S}(U)$  are local semantic threads, claims, cached structures, or repaired routines valid within  $U$ . Morphisms are semantic restrictions or rewrites. For inclusions  $V \subseteq U$ , the restriction functor  $\rho_V^U : \mathcal{S}(U) \rightarrow \mathcal{S}(V)$  encodes context reduction: what is asserted in the larger context may be forgotten or specialized in the smaller one.

## 6.2 The Sheaf Condition and Semantic Repair

### Definition 6.2 (Semantic Tear and Repair)

A family of local sections  $\{s_i \in \mathcal{S}(U_i)\}$  is *glueable* if: (i) local consistency holds:  $\rho_{U_i \cap U_j}^{U_i}(s_i) \cong \rho_{U_i \cap U_j}^{U_j}(s_j)$  for all overlapping pairs; and (ii) there exists a global extension  $s \in \mathcal{S}(\bigcup_i U_i)$  restricting to each  $s_i$ .

A *semantic tear* occurs when local consistency fails on some overlap  $U_i \cap U_j$ : the restriction of  $s_i$  and the restriction of  $s_j$  to their common domain are not isomorphic. A *repair* is the introduction of new objects or morphisms into  $\mathcal{S}(U_i)$  or  $\mathcal{S}(U_j)$  such that glueability is restored.

## 6.3 Cohomology as Semantic Entropy

Cohomology detects the unresolved incompatibilities that remain after local repairs have been applied. The Čech cohomology groups of the presheaf  $\mathcal{S}$  with respect to the cover  $\mathcal{U}$  organize



these obstructions by dimension:

- $H^0(X, \mathcal{S})$  viable global sections (coherent meanings),
- $H^1(X, \mathcal{S})$  minimal ambiguities (semantic seams),
- $H^k(X, \mathcal{S}), k \geq 2$  deeper obstructions to coherence.

These cohomology groups are interpreted as measures of semantic entropy: the unresolved cocycles represent ambiguity stored for potential reinterpretation. Repair corresponds to introducing new morphisms that trivialize cocycles, moving from  $H^1$  toward  $H^0$  by resolving the incompatibilities that the cocycles record.

*Strategic ambiguity is a resource rather than a failure. A Yarncrawler that collapses all ambiguity immediately—forcing a unique global section at every step—forfeits the adaptability that deferred gluing provides. The presence of a non-trivial  $H^1$  is a signal that multiple viable interpretations exist; the system defers collapse until additional evidence arrives to determine which interpretation to commit to.*

## 6.4 The Stigmergic Repair Closure Theorem

### Theorem 6.1 (Stigmergic Repair Closure)

Let  $\mathcal{S}$  be a presheaf of local semantic sections over patches  $\{U_i\}$ , with overlaps  $U_i \cap U_j$  that may admit inconsistent interpretations. Suppose that for each overlap there exists an elastic repair operation  $\rho_{ij}$  satisfying:

1.  $\rho_{ij}$  preserves type safety: objects remain within a coarser semantic category (for example, “seed-like” rather than either “seed” or “pebble”);
2.  $\rho_{ij}$  reduces local free energy  $\mathcal{F}(U_i \cap U_j)$  relative to the unrepaired sections; and
3.  $\rho_{ij}$  is stigmergic: the cost of subsequent repairs decreases with the density of prior contributions.

Then the family of repaired sections  $\{s_i^\rho\}$  admits at least one global section  $s$  with  $\mathcal{F}(s) \leq \sum_i \mathcal{F}(s_i)$  and with viability increasing in the cumulative density of repairs.

*Proposition 6.1 (Repair Preserves Blanket Structure)*

If every repair is generated by local morphisms that commute with restrictions and respect conditional independence at the Markov blanket—that is, if  $X_t \perp\!\!\!\perp E_t \mid (S_t, A_t)$  is maintained throughout—then any finite composition of repairs preserves the gluing structure of  $\mathcal{S}$ .

## 6.5 Self-Maintaining Growth under Stigmatic Feedback

**Theorem 6.2 (Self-Maintaining Growth)**

Let  $\Phi(t)$  denote the accumulated density of a repaired structure (cache, berm, semantic section). Suppose the repair cost function  $C(n)$  for the  $n$ -th contribution is monotonically decreasing in cumulative density  $\Phi(n)$ . Then there exists a critical threshold  $\Phi^*$  such that

$$\Phi(n+1) - \Phi(n) \geq \delta > 0 \quad \text{whenever } \Phi(n) \geq \Phi^*,$$

that is, once  $\Phi^*$  is surpassed, the system enters a regime of self-sustaining growth where each new repair lowers the barrier for future repairs.

The discrete-time dynamics implementing this theorem take

the form:

$$\Phi_{t+1} = \Phi_t + \Delta t \left( \alpha \frac{\Phi_t}{\Phi_t + \theta} \left( 1 - \frac{\Phi_t}{K} \right) - \delta \Phi_t \right),$$

where  $\alpha \equiv \lambda\kappa$  is baseline inflow,  $\theta > 0$  encodes early friction (the cost of initial contributions),  $K$  is carrying capacity, and  $\delta \geq 0$  is decay. The stigmergic threshold condition is  $\alpha > \delta\theta$ .

## 6.6 Ecological and Cultural Instantiations

The abstract sheaf-repair mechanism is grounded by two concrete instantiations that run across the ecological and cultural domains analyzed in the companion work.

In squirrel foraging, each foraging episode is a local patch  $U_i$  with a section  $s_i$  encoding the interpretation of an encountered object. A nut recognized in one context and a visually similar pebble encountered in another produce conflicting restrictions at the overlap. Rather than collapsing, the squirrel repairs by adopting a softened category—“seed-like object.” The cocycle condition fails strictly but is restored under a relaxed gluing criterion, preserving viability. Multiple possible global interpretations are held in suspension until additional evidence resolves the ambiguity.

In berm construction, an analogous process occurs at the cultural scale. A villager encountering a cairn at a territorial edge may perceive it as insufficient and add more material. The mate-

rial composition need not be uniform; bones, ash, and dung may be included alongside rocks. Local repairs introduce semantic elasticity: the criterion is “boundary-marker-like” rather than “stone-only.” Over time, overlapping deposits cohere into self-maintaining structures that attract further contributions, forming the Voronoi-like tessellations of Amazonian terra preta.

In both cases, the global section—the food cache or the territorial berm—emerges not from top-down planning but from the gluing of locally inconsistent but compatible acts of repair. This is the universal mechanism of the Yarncrawler framework: local disagreements reconciled through semantic elasticity, strategic ambiguity, and stigmergic reinforcement.

## Chapter 7

# Ecological, Cultural, and Computational Projections

Across ecological, cultural, and artificial domains, the invariant dynamic is stigmergic reinforcement coupled to Markov blanket repair.

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## 7.1 The Spectrum from Squirrel to Language Model

The Yarncrawler principle is that agents do not merely consume; they weave, cache, and repair. Each action leaves a trace that both satisfies immediate needs and sets the stage for recursive self-maintenance. This dynamic recurs across ecological, cultural, and computational scales. The invariants under this scale change are the `rsvp` field correspondences:  $\Phi$  measures accumulated deposits,  $\mathbf{v}$  encodes recursive flows, and  $S$  quantifies robustness.

## 7.2 Ecological Yarncrawlers

A squirrel does not carry a metric map of its forest. Its survival depends on weighted affordances: gradients of safety, concealment, and food density blurred across its perceptual field. When hungry, it follows Gaussian-blurred gradients toward likely caches rather than performing global optimization. Its over-provisioning of seeds is not an error but the creation of semantic attractors—distributed caches that stabilize its future viability.

The perceived affordance field is  $\psi(x) = (G_\sigma * u)(x)$  where  $u(x)$  is the distribution of hidden resources and  $G_\sigma$  is a Gaussian smoothing kernel. Foraging dynamics are  $\dot{x}(t) = -\nabla\psi(x) + \eta(t)$ . Each cache deposit increments  $u(x_k)$ , making the field the squirrel uses simultaneously a navigation tool and a stigmergic memory. The forest itself becomes a distributed memory field, with excess ensuring robustness against local failure.

## 7.3 Cultural Yarncrawlers

Cultural Yarncrawlers instantiate the same principles at a larger scale. In Amazonian communities, household waste was exported to boundary piles that grew fertile through microbial and chemical processes. Once above a critical threshold, these berms self-maintained, attracting further deposition and form-

ing Voronoi-like tessellations of territory. The resulting terra preta soils persist for millennia.

The governing dynamics are analyzed in the companion paper *Yarncrawler in Action* through a system of reaction-advection-diffusion equations for berm biomass  $B$ , char matrix  $C$ , and nutrients  $N$ . The stigmergic threshold condition  $\alpha_B \langle S \rangle_{A_r} > \delta_B \theta_{\text{eff}}$  formalizes the transition from fragile deposition to self-maintaining growth.

## 7.4 Artificial Yarncrawlers

Mixture-of-Experts architectures and retrieval-augmented generation instantiate the Yarncrawler principle computationally. Each expert module functions as a local chart on the semantic manifold. Retrieval mechanisms act as stigmergic glue: a memory trace that is frequently retrieved becomes more salient, just as a cairn or berm attracts additional reinforcement. The global semantic field is assembled from local expert densities by  $\Phi(x) = \sum_i w_i(x) \varphi_i(x)$ .

The key distinction from standard MoE/RAG architectures is that the repair mechanisms of a Yarncrawler reconfigure the modules themselves rather than only gating access to them. The semantic structure is continuous and trajectory-based, modeled by sheaf gluing and RSVP fields, rather than discrete or static.



## 7.5 The Climate Response as Worked Example

The read-evaluate-write-move loop provides the clearest operational instantiation of the `KES` map  $\Omega_t \rightarrow H_{t+1}$  anywhere in the corpus. Consider a Yarncrawler deployed across a coastal region:

**Node 1 (Seattle):** Reads rising sea levels and failing levees. Evaluates that current barriers are underfunded. Writes a plan to redirect local taxes to infrastructure. Moves to Node 2, carrying the tax-redirection strategy as a candidate fix for other urban nodes.

**Node 2 (Olympic National Forest):** Reads increased wildfire risk. Evaluates that invasive species are worsening fire behavior. Writes a schedule for controlled burns and native replanting. Moves to Node 3.

**Node 3 (Portland):** Reads heatwave data. Evaluates that cooling centers are overcrowded. Writes a policy to expand public cooling spaces, using lessons from Seattle’s resource allocation. Moves to Node 4.

The Yarncrawler learns that tax redirection from Seattle outperforms Portland’s grant-based model and applies this to future urban nodes. It notes that controlled burns require longer monitoring cycles, refining its repair strategy for other forest nodes. The trail of fixes constitutes a semantic ledger of the reasoning trajectory—what the companion framework calls Chain of Mem-

ory.

# **Part IV**

## **Operational Projections**

Cognition, computation, physical evolution, institutional dynamics, and representational systems are all instances of the same underlying architecture: constrained transformation through admissible trajectory spaces under thermodynamic, logical, geometric, and historical pressures.

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Part IV applies the Yarncrawler-grounded framework across three domains: cosmology, social and institutional systems, and computation. Each chapter functions as a coordinate chart over the shared geometry established in Parts I–III. The same invariants—admissibility, trajectory selection, compression, reconstruction, irreversibility—resolve differently across substrates while remaining formally recognizable as instances of the same architecture.

## Chapter 8

# Cosmological Projection: RSVP and the Persistence of Structure

Reality is modeled as an ongoing process of constrained transformation rather than a fixed geometric object in which all events coexist.

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### 8.1 Against the Block Universe

Static block-universe interpretations of general relativity treat the universe as a completed four-dimensional object in which all events coexist simultaneously in the tenseless present of space-time geometry. The `rsvp` framework provides a dynamically consistent alternative grounded in irreversibility, entropy gradients, and locally constrained field relaxation.

On the `rsvp` interpretation, the universe is modeled less as a completed four-dimensional object and more as an ongoing pro-

cess of constrained transformation in which present configurations continuously reorganize into future configurations through local interactions, thermodynamic pressures, and evolving accessibility conditions. The entropy field  $S(\mathbf{x}, t)$  is not a static property of spacetime but a dynamical quantity whose gradients drive the evolution of the other fields.

## 8.2 The RSVP Lagrangian and Admissibility Geometry

The full Lagrangian structure is developed in Appendix 9.4. The seven governing axioms specify the coupling between the scalar density field  $\Phi$ , the vector flow field  $\mathbf{v}$ , and the entropy field  $S$ . Admissible cosmological trajectories are those satisfying the derived compatibility conditions across all three fields simultaneously.

### Redshift: Expansion vs. Entropic Relaxation

The standard  $\Lambda$ CDM interpretation of cosmological redshift attributes the wavelength shift of photons to metric expansion: as the universe expands, the physical wavelength of a photon increases in proportion to the scale factor  $a(t)$ , giving  $z = a(t_{\text{obs}})/a(t_{\text{emit}}) - 1$ .

The RSVP interpretation attributes redshift to entropic differ-

ential: photons traversing regions of entropy gradient transfer energy to the  $S$  field rather than to metric stretching. The predicted redshift for a photon traveling through an entropy gradient  $\nabla S$  over path length  $L$  is approximately:

$$z_{\text{RSVP}} \approx \int_0^L \kappa |\nabla S(\ell)| d\ell,$$

where  $\kappa$  is a coupling constant relating entropy gradient to energy loss.

The two interpretations produce observationally similar predictions at low redshift ( $z \lesssim 1$ ) because, in the RSVP framework, the entropy field evolves at rates that mimic scale-factor growth over cosmic time. They diverge at high redshift and at the level of structure formation:  $\Lambda$ CDM predicts that the power spectrum of density fluctuations is seeded by inflation and modulated by dark matter, while RSVP predicts that structure forms through entropy gradient focusing and scalar field condensation without requiring either an inflationary epoch or exotic dark matter. The RSVP framework therefore makes empirically distinct predictions about the matter power spectrum at  $k \gtrsim 0.1 \text{ h/Mpc}$  and about the Silk damping scale, which constitute its primary falsification criteria. Cosmological redshift, on this interpretation, arises from entropic differential rather than metric expansion: photons traversing regions of entropy gradient lose energy to the gradient field rather than to the stretching of space. This generates observationally similar predictions to expansion cosmology in

the low-redshift regime while making distinct predictions at the structure formation scale.

## 8.3 Persistence, Residue, and Structural Memory

Physical systems do not merely evolve forward through admissible trajectories. They accumulate structural residue: traces of prior constraint configurations that continue influencing future evolution. Fossil manifolds—large-scale structures whose current configuration encodes the constraint history of their formation—are cosmological counterparts to the provenance concept introduced in Chapter 2.

### Definition 8.1 (Structural Residue)

The *structural residue* of a dynamical process over interval  $[t_0, t]$  is the component of the current state  $\Omega_t$  that is causally attributable to constraint configurations that no longer operate at  $t$  but that produced persistent modifications to the state space topology during  $[t_0, t]$ . Structural residue is the physical instantiation of provenance: the history of the system is encoded in its present configuration in a form that, in principle, supports reconstruction of prior states.



## **Part V**

# **Method as Consequence**

The writing style is a mechanistic consequence of the theory, not a stylistic preference. A text that argues for provenance preservation while itself operating through aggressive compression would be internally inconsistent.

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Part V makes explicit what has been enacted throughout the preceding parts: the writing philosophy, epistemological substrate, and self-referential structure of this monograph are mechanistic entailments of the theory rather than stylistic preferences. The conclusion draws the Juarrero–Deacon distinction on teleology and states the process-ontological thesis of the monograph in its most direct form.

## Chapter 9

# Emergent Directedness Without Inscribed Teleology

Meaning is not pre-inscribed into the universe. It is progressively assembled through persistent selective dynamics operating across scales.

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### 9.1 The Juarrero–Deacon Distinction

Alicia Juarrero's account of context-sensitive dynamical constraints and Terrence Deacon's analysis of absence, constraint, and emergent organization both influence the framework developed in this monograph, but neither is adopted wholesale. Deacon's concept of absential organization sometimes approaches the claim that intentionality is ontologically inscribed into nature itself. This section articulates the distinction between that position and the process ontology developed here.

Purposive behavior, apparent goal-directedness, and semantic organization are treated in this framework as emergent consequences of layered reconstruction processes navigating constrained state spaces under irreversible historical conditions. Meaning is not pre-inscribed. It is progressively assembled through persistent selective dynamics operating across scales.

## **Derivation: Apparent Purposiveness from Irreversible Constraint Accumulation**

The distinction between emergent directedness and inscribed teleology can be made precise. Consider a system evolving under irreversible constraint accumulation: at each step  $t$ , a new constraint  $c_t$  is added to the admissible set  $\mathcal{C}_t$ , with  $\mathcal{C}_{t+1} = \mathcal{C}_t \cup \{c_{t+1}\}$ .

Since constraints only ever contract the admissible trajectory space (as shown in the Chapter 1 derivation), the sequence  $\mathcal{A}(\mathcal{C}_0) \supseteq \mathcal{A}(\mathcal{C}_1) \supseteq \mathcal{A}(\mathcal{C}_2) \supseteq \dots$  is a decreasing filtration. Over time, the system's accessible future narrows. An observer watching from outside this process will observe the system appearing to "aim at" the configurations that survive all constraints: those configurations that have not been ruled out by any  $c_t$ .

This appearance of directedness requires no pre-inscribed goal. It follows structurally from the irreversibility of constraint accumulation: the system cannot revisit ruled-out trajectories, so its motion in state space is asymmetric, exhibiting a directionality that did not exist before the constraints were accumulated. Ap-

parent purposiveness is the phenotype of irreversible constraint history.

The crucial difference from Deacon's absential account is that this framework does not require positing an attracting absence—a not-yet-realized state that exercises causal influence on present dynamics. The directionality is entirely explained by the accumulated record of what has been ruled out. It is backward-looking (in the sense of being generated by prior constraints) rather than forward-looking (in the sense of being pulled toward a pre-existing target).

## 9.2 The Unified Architecture Restated

Cognition, computation, physical evolution, institutional dynamics, and representational systems are all instances of the same underlying architecture: constrained transformation through admissible trajectory spaces under thermodynamic, logical, geometric, and historical pressures, with structural residue accumulating across transformations and reconstruction operating continuously from partial traces.

The Yarncrawler is the self-repair layer of this architecture. The `rsvp` field system is its physical instantiation. The `kes` map is its cognitive operationalization. `Spherepop` is its event-calculus formalization. `TARTAN` is its geometric interface. `CLIO` is its constraint-leveraged inference mechanism. These are not separate frame-

works that have been forced into artificial unity. They are coordinate charts over the same manifold, compatible on their overlaps, and jointly covering the space of complex adaptive systems under constraint.

## **9.3 Implications and Open Trajectories**

The framework generates specific empirical and design implications. Systems should be designed for corrigibility rather than seamlessness. Representational infrastructure should preserve provenance rather than optimizing for compression. Institutions should be evaluated against their topological admissibility conditions rather than only their stated objectives. Education should prioritize boot-sequence reconstruction over terminal-state memorization.

## **9.4 The Remaining Challenge**

Transforming a sprawling recursive semantic ecosystem into a navigable object without destroying the generative richness that produced it. That is the problem this monograph enacts rather than merely describes. Whether the constraint geometry imposed here preserves enough of the original manifold's topology to remain useful is a question only the reader's reconstruction can answer.

# Formal Specification of the RSVP Field System

Formal specification to be populated from companion documents including *Axioms for a Falling Universe* and *Geometry from Entropy Flow*. The seven axioms governing field evolution and the full Lagrangian structure will be reproduced here.

## The KES Map: Formal Specification

Formal specification of the Kinetic-Event Synthesis map  $\Omega_t \rightarrow H_{t+1}$ , including convergence conditions and the 56-page treatment developed in the companion KES convergence monograph.



## **Spherepop Operator Semantics**

Operational semantics for the four Spherepop operators: Pop, Refuse, Bind, and Collapse. Worked examples and relationship to Spherepop game engine implementation.

# **TARTAN: Trajectory-Aware Recursive Tiling with Annotated Noise**

Architecture specification for the TARTAN framework, including the annotated noise schema, four projection modes, and relationship to hallucination formalism.

# **CLIO: Constraint-Leveraged Inference and Optimization**

Specification of the CLIO recursive constraint repair framework, including its relationship to the Cheng, Broadbent, and Chappell (2025) system and the distinctions developed in Chapter 5.

# **Obstruction Cohomology and Katz-Admissibility**

Mathematical background for the sheaf-theoretic institutional failure analysis of Chapter ?? . Katz-admissibility conditions and global extension theorems.

## **Juarrero and Deacon: Selected Framework Comparisons**

Comparative analysis of context-sensitive dynamical constraints (Juarrero 1999) and absential organization (Deacon 2011) against the process ontology developed in this monograph.

# The Stone Piano Hypothesis: An Archaeological Case Study

Whatever they were doing, they intended it.

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The Stone Piano Hypothesis proposes that the Bruniquel Cave stalagmite structures, dated to approximately 176,500 years before present and attributed to *Homo neanderthalensis*, functioned as lithophones: percussion instruments fabricated from resonant calcite speleothems. This appendix situates the hypothesis within the Yarncrawler framework.

## 9.5 Admissibility Filtering in Speleothem Selection

The fundamental frequency of a free-free calcite rod of length  $L$  and radius  $r$ :

$$f_1 = \frac{\lambda_1^2 r}{8\pi L^2} \sqrt{\frac{E}{\rho}}, \quad \lambda_1 \approx 4.730,$$

with  $E \approx 75 \text{ GPa}$  and  $\rho \approx 2600 \text{ kg m}^{-3}$ . Fracturing stalagmites from their bases converts clamped-free resonators (inharmonic series 1 : 6.27 : 17.55) into free-free resonators with more harmonic profiles (1 : 2.76 : 5.40). This is an admissibility filtering operation.

## 9.6 Sympathetic Resonance as the Corrigibility Mechanism

Acoustic beating—the periodic amplitude modulation produced by two nearly coincident frequencies—disappears when frequencies are matched. Its disappearance is an unambiguous perceptual event providing real-time corrigibility feedback. The builders could tune fragment lengths by progressive shortening, without abstract acoustic knowledge, using only the disappearance of beating as a signal. This is the *KES* loop operating in stone at 176,500 BP.

## 9.7 The Ring as Stigmergic Structure

A partially assembled ring of resonant elements constitutes an acoustic admissibility filter. New candidate fragments that excite strong sympathetic vibrations in the existing ring are harmonically admissible; those that do not are refused. The structure under construction participates in selecting the components from

which it is built: Axiom 3 (Autocatalytic Fix Design) instantiated in deep prehistory.

## 9.8 Estimated Chamber Reverberation

By Sabine's formula with  $V \approx 73 \text{ m}^3$  and  $\alpha \approx 0.03$ :

$$T_{60} \approx \frac{0.161 \times 73}{0.03 \times 130} \approx 3.0 \text{ s.}$$

The cave amplifies and extends each repair operation, lowering the threshold for subsequent ones. It is a Yarncrawler environment.



## Companion Document Index

The following companion documents are cross-referenced throughout this monograph. All are available at <https://github.com/standa>

**Physics and Cosmology:** *Axioms for a Falling Universe; The Persistence of Structure; Admissible Geometry; Fossil Manifolds and Mediated Extraction; Geometry from Entropy Flow; Physics After Spacetime; Scalar Irruption via Entropic Differential.*

**Cognition and Epistemology:** *The Collapse of Epistemic Coherence; Hallucination is Normal; Never Predict Noise; World-State Reconstruction; Constraint Before Completion; Selection Without Design; The Compiled Self; Noun-Free Cognition.*

**Computation:** *Yarncrawler in Action; The Elements of Computational Worlds; Geometry, Cognition, and the Transparency of Computation; Semantic Relaxation Networks; Trajectories All the Way Down; Irreversibility as Architecture.*

**Social and Institutional:** *The Collapse of Proxy Integrity; Participation without Guarantee; Recomposable Fragmentation; The Entropy of Austerity; Obstruction Cohomology; Against Namespace Laundering; The Stack Capture Race.*

**Paleoarchaeology:** *The Stone Piano Hypothesis.*

## **Bibliography**